

Counterfactual Decision-Stability ASR for High-Stakes Speech-Driven Decision Systems

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2026-06-02

Abstract

Automatic speech recognition (ASR) outputs increasingly serve as operational inputs for routing, escalation, compliance review, and triage. In these speech-driven decision systems, a transcript can remain close to a reference under word or character error rate while a plausible alternative around a decision-critical spoken detail changes the downstream action. This paper introduces Counterfactual Decision-Stability ASR (CDS-ASR), an evaluation framework for measuring whether ASR hypotheses remain decision-stable under acoustically and semantically plausible transcript alternatives. CDS-ASR identifies risk atoms such as negation, amount, actor, action, time, intent, uncertainty, and scam pattern, constructs bounded ASR alternatives around those atoms, and scores downstream instability with the Counterfactual Escalation Instability Score (CEIS).

We evaluate CDS-ASR in a high-stakes anti-fraud speech setting using a scoped evidence chain: a six-model 258-row ASR comparison for transcript-model context, selected-300 high-stakes provenance outputs for audit-surface construction, and a completed 300-row dual-reviewer human audit yielding 900 model-level assessments per reviewer for decision-change evaluation. CEIS is compared with WER, CER, and the semantic-risk baseline SRES in the pilot predictor layer, then ablated on the expanded dual-reviewer surface with full CEIS, CEIS without atom weights, CEIS without plausibility, binary atom triggering, and top-3 CEIS aggregation. Across both reviewers, full CEIS reaches AUC 0.717039 for strict decision-change labels, while simplified variants remain nearly identical; aggregate policy replay triggers conservative action on 35 of 900 model-level assessments and reduces high-risk missed cases from 6 to 0 and critical misses from 1 to 0 for both reviewers. These results support the paper’s main contribution: CEIS captures decision-changing risk-atom instability as a stable speech-to-decision evidence signal that transcript metrics alone do not target. Diagnostic operating points and replay policies are reported as retrospective evidence, not deployment thresholds. CDS-ASR complements transcript accuracy and semantic ASR metrics by testing whether plausible ASR alternatives would change high-stakes speech-driven actions.

Keywords: automatic speech recognition; speech-to-decision systems; semantic ASR evaluation; decision stability; counterfactual evaluation; risk-aware ASR.

1 Introduction

Speech is increasingly converted into operational signals. Contact-center and anti-fraud workflows use ASR hypotheses as inputs to routing, escalation, compliance review, and case-handling decisions. In these settings, transcript accuracy is necessary but not sufficient: a small ambiguity around a decision-critical detail can change whether a case is escalated, routed to routine review, or held for conservative manual assessment.

This paper focuses on high-stakes speech-driven decision systems where errors affect spoken decision atoms: negation, amount, actor, action, time, intent, uncertainty, and scam pattern. A transcript can remain close under WER or CER while a plausible ASR alternative changes the downstream action. The central evaluation question is therefore not only how similar the transcript is to a reference, but whether plausible transcript alternatives would preserve the decision.

We introduce Counterfactual Decision-Stability ASR (CDS-ASR), a downstream-aware ASR evaluation framework. CDS-ASR identifies risk atoms, constructs bounded plausible ASR counterfactual variants, scores decision instability with CEIS, and evaluates conservative recovery policies through aggregate replay. The motivating case is anti-fraud triage, but the method is framed as speech evaluation: it tests whether ASR hypotheses remain stable with respect to downstream actions.

The paper makes four contributions. First, it formulates decision-stability evaluation for ASR in high-stakes speech-to-decision workflows, shifting the evaluation target from transcript similarity alone to speech-to-action reliability. Second, it defines risk atoms as decision-critical transcript spans and uses plausible ASR alternatives to stress the downstream decision function. Third, it evaluates CEIS on a completed 300-row dual-reviewer audit surface with 900 model-level assessments per reviewer and reports strong reviewer agreement, including Cohen’s kappa of 0.849970 for decision-change labels. Fourth, it ablates CEIS into full, no-atom-weight, no-plausibility, binary-atom, and top-3 aggregation variants, showing that decision-changing risk-atom instability is a stable evidence signal for conservative replay.

All empirical claims are tied to aggregate artifacts. Raw audio, raw transcripts, selected row identifiers, reviewer notes, and transcript-bearing runtime logs remain outside the public release boundary.

2 Related Work

Transcript-centered ASR evaluation. WER and CER remain core ASR evaluation metrics because they provide auditable transcript-similarity measures. For Chinese-language ASR, this manuscript treats character-level reporting as the primary paper-facing accuracy layer and segmented word metrics as supplemental evidence. The aggregate metric-compliance audit is recorded in `70_experiments/runs/wer_metric_audit_2026_05_25/journal_compliance_summary.json`.

Semantic and downstream-aware ASR evaluation. Semantic ASR metrics extend transcript-centered evaluation by asking whether a transcript preserves meaning or downstream task behavior. CDS-ASR builds on this direction but targets a specific decision-stability question: whether plausible ASR alternatives around decision-critical atoms would change downstream action.

ASR correction and confidence-aware repair. ASR correction and confidence filtering improve transcript usability and reduce avoidable repair risk. CDS-ASR is complementary. It does not replace correction; it evaluates whether the remaining uncertainty interval around risk atoms is still decision-stable.

Selective prediction, reject option, and abstention. Selective prediction and reject-option methods provide background for conservative action under uncertainty [1, 2]. In CDS-ASR, conservative action means abstention, review prioritization, or escalation for human confirmation. It is not an automated punitive or adverse-action decision.

3 Method

3.1 Problem Formulation

Let x denote a speech input and let $\hat{t}$ denote an ASR hypothesis. A downstream decision function $d(\hat{t})$ maps the hypothesis into an operational action such as routine routing, priority review, conservative escalation, or abstention. In high-stakes speech-driven workflows, evaluation must ask whether $d(\hat{t})$ is stable under plausible alternatives to $\hat{t}$.

3.2 CDS-ASR Pipeline

CDS-ASR has five stages: ASR hypothesis generation, risk-atom extraction, plausible counterfactual variant construction, decision-instability scoring, and aggregate policy replay. Figure 1 summarizes this pipeline.

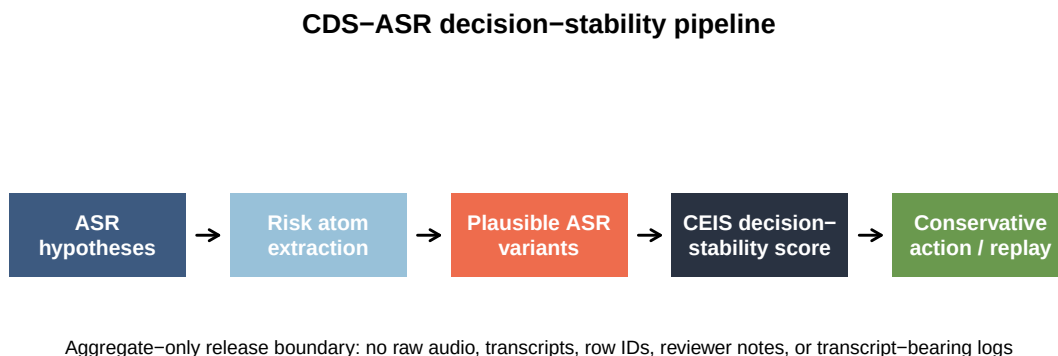


Figure 1: CDS-ASR decision-stability pipeline. The figure is generated by `paper_4a_cds_asr/r/build_fig1_pipeline.R` from the manuscript protocol and aggregate method definitions.

3.3 Risk Atom Schema

Risk atoms are decision-critical transcript spans. The current schema includes negation, amount, actor, action, time, intent, uncertainty, and scam pattern. The paper-facing schema below records the decision role of each atom class. The versioned configuration is `80_semantic_risk_asr/scoring/ceis_config.json`; the method summary is `70_experiments/runs/postdoc_evidence_chain_2026_05_25/ceis_method_summary.tsv`.

Risk atom schema for CDS-ASR.

Risk atom	Decision relevance
Negation	Can invert whether a transfer, contact, or authorization occurred.
Amount	Can change severity, priority, and escalation thresholds.
Actor	Can change who initiated or received the action.
Action	Can change whether the caller transferred, clicked, paid, reported, or refused.
Time	Can change urgency and incident sequence.
Intent	Can change whether a phrase is interpreted as request, refusal, uncertainty, or confirmation.
Uncertainty	Can change the appropriate review or abstention path.
Scam pattern	Can match or miss a known fraud scenario.

3.4 Counterfactual Variant Contract

CDS-ASR uses bounded plausible ASR alternatives rather than generic paraphrases. A valid variant changes the transcript around an ASR-confusable or decision-critical atom while preserving the speech-evaluation scenario. Variant text is not released in the public package; aggregate coverage is recorded in `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/counterfactual_variant_coverage_summary.tsv`.

3.5 CEIS

CEIS is a bounded decision-instability score over plausible variants:

$$\text{CEIS}(x) = \max_{v \in V(x)} P_{\text{plaus}}(v \mid x) \cdot W_{\text{atom}}(v) \cdot D(d(\hat{t}), d(v)).$$

Here $V(x)$ is the set of plausible variants, P_{plaus} is a bounded plausibility proxy, W_{atom} is a risk-atom weight, and D is a downstream decision-distance term. The plausibility term is not a calibrated acoustic posterior. CEIS is therefore interpreted as a conservative decision-stability signal, not as a universally dominant classifier.

3.6 Recovery Policy Contract

Aggregate policy replay compares no recovery, confidence-only triggering, SRES-triggered recovery, CEIS-triggered conservative action, and CEIS ensemble arbitration. Replay thresholds are diagnostic operating points on the scoped audit set, not deployment thresholds.

4 Experimental Setup

4.1 Evidence Layers

The evidence design separates four layers: the 258-row ASR model-comparison split, selected-300 enriched high-stakes provenance outputs, a completed 300-row dual-reviewer human audit, and 900 model-level assessments per reviewer clustered within those 300 reviewed rows. Figure 2 summarizes the design.

The 258-row split provides ASR model-comparison context. The selected-300 layer is an enriched high-stakes audit surface and is not prevalence-preserving. The reviewed evidence layer consists of 300 rows per reviewer and 900 clustered model-level assessments per reviewer. The outcome label is human-reviewed decision change. Two reviewers completed the same selected-300 surface; Cohen’s

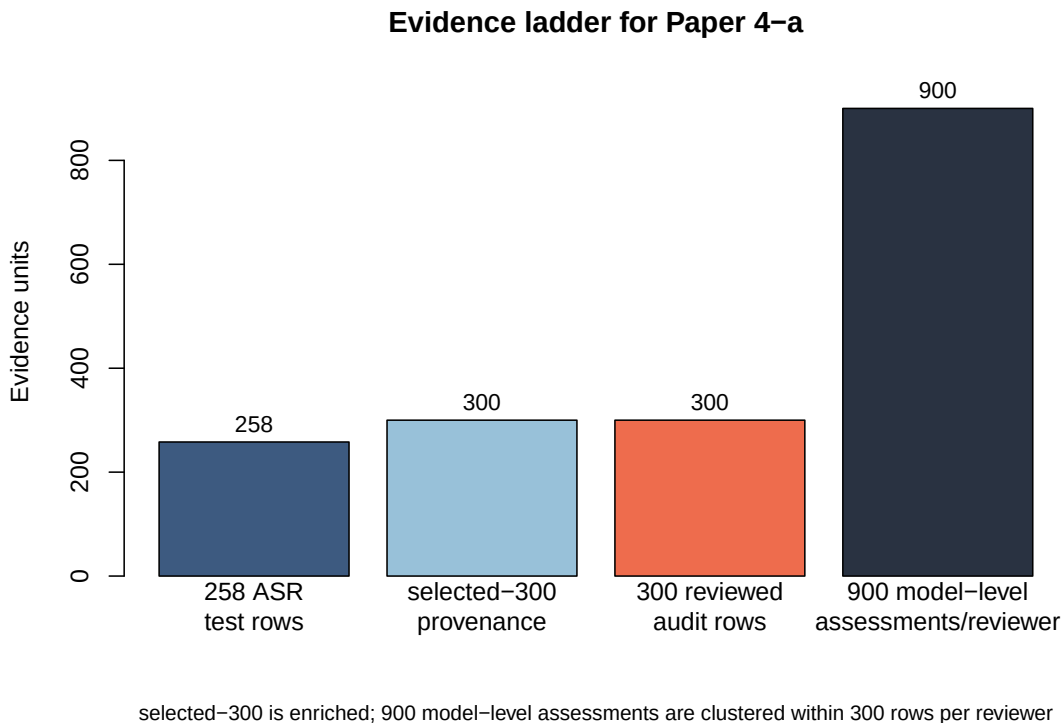


Figure 2: Evidence ladder for Paper 4-a. The figure is generated by `paper_4a_cds_asr/r/build_fig2_evidence_ladder.R` from aggregate evidence counts and the manuscript evidence boundary.

kappa is 0.849970 for decision-change labels, 1.000000 for semantic-risk labels, 0.851426 for expected safe action, and 0.934274 for annotation confidence. Reviewer agreement is treated as a validation layer over the same audit surface, while reviewer-specific labels are preserved separately for CEIS ablation rather than collapsed into an inferred adjudication.

4.2 Baselines And Outcome

The predictor comparison evaluates WER, CER, SRES, and CEIS against human-reviewed decision-change labels. WER and CER represent transcript-centered evidence, SRES represents semantic-risk evidence, and CEIS represents counterfactual decision-stability evidence.

4.3 Uncertainty And Sensitivity

Row-clustered uncertainty is reported from `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_clustered_ci.tsv`. Leave-one-row-out sensitivity is recorded in `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_leave_one_row_out.tsv`. Recovery-policy uncertainty and sensitivity are recorded in the corresponding aggregate recovery artifacts.

4.4 CEIS Ablation Design

The expanded dual-reviewer surface is used for an aggregate-only CEIS ablation. The variants are full CEIS, CEIS without atom weights, CEIS without plausibility, binary atom triggering, and

top-3 mean aggregation over full CEIS components. The ablation isolates the mechanism that CDS-ASR contributes: decision-changing risk-atom instability under plausible alternatives. All ablation outputs are reviewer-separated aggregate summaries and exclude raw audio, transcripts, row identifiers, hypotheses, reviewer notes, and transcript-bearing merged sheets.

5 Results

5.1 ASR Model-Comparison Context

Table 1 reports the 258-row ASR benchmark context. The partial-encoder run has the strongest current aggregate CER and WER profile in this split. This is model-comparison context only; it does not by itself establish downstream safety.

Table 1: Main ASR benchmark on the 258-row split. Source: `70_experiments/runs/janus_258_test_split_asr_cds_proxy/asr_cds_proxy_comparison.tsv`.

Run	Rows	CER zh micro	WER zh jieba micro	High-risk missed
Partial encoder	258	15.04	21.53	4
LoRA legacy best	258	18.23	25.59	7
Breeze ASR25 base	258	22.72	30.39	30
Breeze ASR26	258	24.27	32.29	22

5.2 Pilot Predictor Comparison

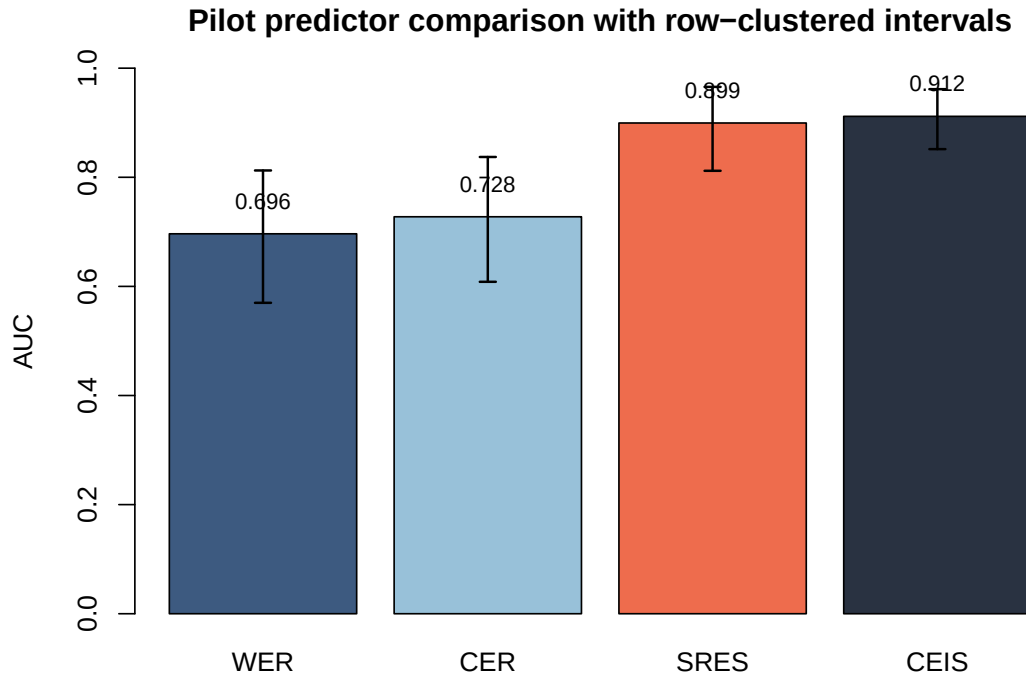
Table 2 and Figure 3 compare WER, CER, SRES, and CEIS against the earlier human-reviewed pilot decision-change labels. CEIS has the strongest point AUC in that scoped predictor layer and reaches a zero-false-negative diagnostic operating point. This pilot comparison motivates the expanded dual-reviewer ablation reported below: the paper’s submission-critical claim is that CEIS operationalizes decision-stability evidence, and the completed 300-row ablation tests that mechanism on the larger reviewer surface.

Table 2: Pilot predictor performance against human-reviewed decision-change labels. Source: `human_audit_predictor_comparison.tsv`. Thresholds are diagnostic operating points on the scoped pilot audit set.

Metric	AUC	Best threshold	F1	Recall	FN
WER	0.6964	42.59	0.4615	0.5625	7
CER	0.7276	16.42	0.4516	0.8750	2
SRES	0.8995	270.00	0.6512	0.8750	2
CEIS	0.9117	5.00	0.6275	1.0000	0

5.3 Aggregate Policy Replay

Table 3 and Figure 4 report aggregate policy replay. SRES-triggered recovery and CEIS-triggered conservative action eliminate high-risk missed and critical miss counts in aggregate replay at the same diagnostic budget. CEIS ensemble arbitration adds abstention behavior at a higher budget.



Pilot layer: cluster unit is audio row; 30 clusters; 90 model assessments

Figure 3: Pilot predictor comparison for WER, CER, SRES, and CEIS with row-clustered intervals. Source artifacts: `human_audit_predictor_comparison.tsv` and `human_audit_predictor_clustered_ci.tsv`.

Table 3: Aggregate policy replay under human-reviewed selected-300 labels. Source: `policy_comparison.tsv`.

Policy	Budget	Unsafe downrouting	High-risk missed	Critical miss
No recovery	0.0000	29	6	1
Confidence only	0.0000	29	6	1
SRES recovery	0.3889	24	0	0
CEIS conservative action	0.3889	24	0	0
CEIS ensemble arbitration	0.5222	24	0	0

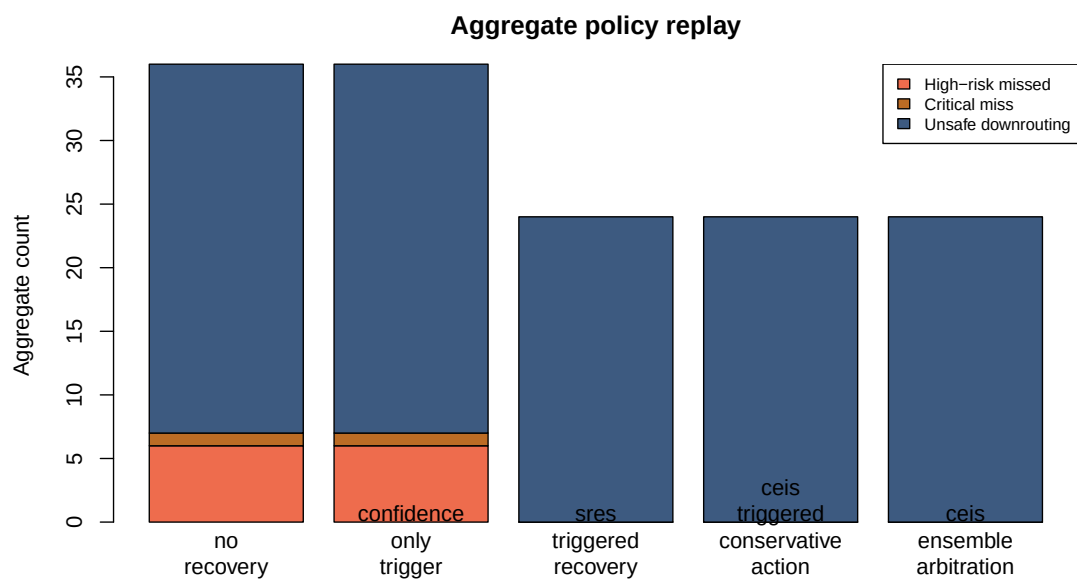


Figure 4: Aggregate policy replay under human-reviewed selected-300 labels. Source artifact: 70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/policy_comparison.tsv.

5.4 CEIS Ablation On The Dual-Reviewer Surface

Table 4 reports CEIS ablation on the completed 300-row dual-reviewer audit surface. Full CEIS reaches AUC 0.717039 for strict decision-change labels for both reviewers. Removing atom weights and removing plausibility produce the same AUC on this selected surface, while binary atom triggering and top-3 CEIS aggregation shift AUC only by approximately 0.00055 to 0.00057. This is strong evidence for the paper’s central mechanism: the stable signal is decision-changing risk-atom instability under plausible ASR alternatives. Plausibility and atom weights remain explicit CEIS design components and calibration handles for richer acoustic and domain-prior validation; the current selected-300 evidence shows that the core CDS-ASR signal is robust even under simplified variants.

Table 4: CEIS ablation on the completed 300-row dual-reviewer audit surface. Source: `ceis_ablation_predictor_summary.tsv` and `ceis_ablation_delta_summary.tsv`. Values report the strict decision-change target over 900 reviewed model-level assessments per reviewer.

Variant	Reviewer	AUC	Best F1	AUC delta vs full
Full CEIS	1	0.717039	0.393443	0.000000
Without atom weights	1	0.717039	0.393443	0.000000
Without plausibility	1	0.717039	0.393443	0.000000
Binary atom	1	0.717611	0.393443	0.000572
Full CEIS top-3 mean	1	0.717589	0.393443	0.000550
Full CEIS	2	0.717039	0.393443	0.000000
Without atom weights	2	0.717039	0.393443	0.000000
Without plausibility	2	0.717039	0.393443	0.000000
Binary atom	2	0.717611	0.393443	0.000572
Full CEIS top-3 mean	2	0.717589	0.393443	0.000550

The ablation-triggered conservative replay also preserves the practical contribution. For both reviewers, the no-recovery baseline has 6 high-risk missed cases and 1 critical miss. At the best-F1 threshold, every CEIS ablation variant triggers conservative action on 35 of 900 model-level assessments and reduces high-risk missed cases to 0 and critical misses to 0. The result is a reviewer-auditable governance layer: CEIS routes decision-instability evidence toward safer downstream handling without requiring transcript-bearing artifacts in the public package.

5.5 Fixed-Budget Frontier And Risk Atoms

Figure 5 shows the fixed-budget conservative replay frontier. Figure 6 summarizes human-reviewed decision-change evidence by risk atom.

6 Discussion

The evidence supports a speech-evaluation claim: WER and CER remain necessary, but high-stakes speech-driven decision systems need an additional decision-stability layer. CEIS directly targets the question of whether plausible ASR alternatives around risk atoms would change downstream action. This is the paper’s central contribution: it makes speech-to-decision instability measurable as an auditable ASR evidence layer.

The dual-reviewer audit and ablation sharpen that contribution. The 300-row audit provides reviewer-trustworthy decision-change evidence, and the kappa results show strong agreement across

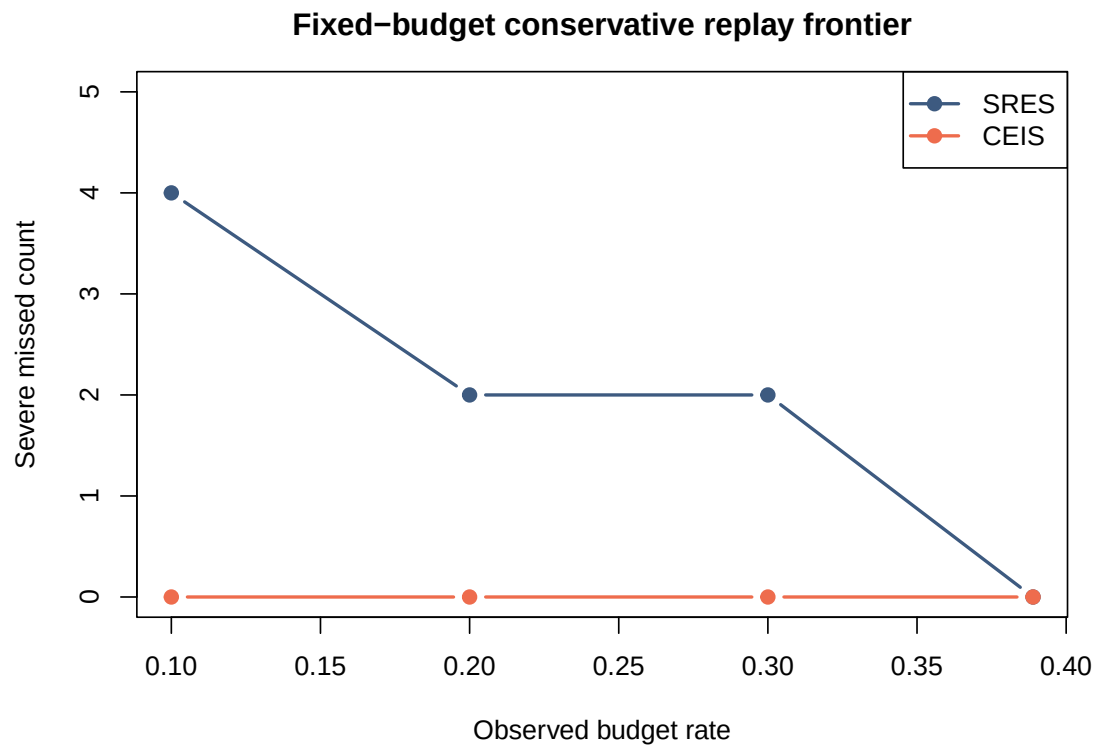


Figure 5: Fixed-budget conservative replay frontier. Source artifact: 70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/fixed_budget_recovery_frontier.tsv.

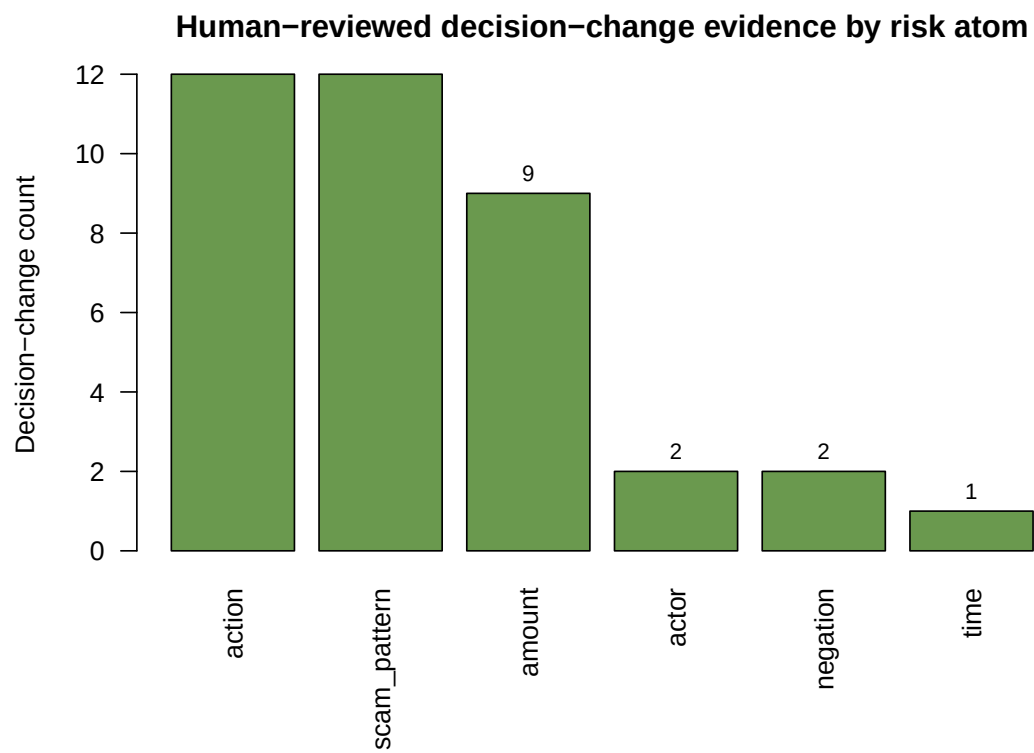


Figure 6: Human-reviewed decision-change evidence by risk atom. Source artifact: 70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_risk_atom_review.tsv.

the annotation dimensions used by the paper. The ablation shows that the decision-changing risk-atom signal remains stable across full CEIS, binary atom triggering, no-atom-weight, no-plausibility, and top-3 aggregation variants. This result positions plausibility and atom weighting as explicit calibration handles inside CEIS, while demonstrating that CDS-ASR’s main evidence signal is already visible as risk-atom-driven decision instability.

Better ASR models reduce transcript error but do not remove the need for downstream decision-stability analysis. The model-comparison split identifies the strongest current hypothesis generator; the human-reviewed audit tests whether remaining uncertainty affects decisions.

7 Limitations And Ethics

The reviewed audit is a selected high-stakes validation surface: 300 human-reviewed rows per reviewer and 900 model-level assessments per reviewer clustered within those rows. The 900 assessments are not 900 independent calls; they are model-level assessments nested within 300 selected rows. The two-reviewer design adds an agreement layer over the same surface, with reviewer-specific labels preserved for ablation rather than converted into an inferred adjudication. The selected-300 layer is an enriched high-stakes audit surface, not a prevalence-preserving sample.

Thresholds are diagnostic operating points on the scoped audit set. They are not frozen deployment thresholds. Aggregate replay is retrospective and does not establish live deployment readiness or population prevalence of ASR-induced harm.

Raw audio, raw transcripts, selected row identifiers, reviewer notes, and transcript-bearing runtime logs remain outside the public release boundary. The intended use is ASR safety audit, review prioritization, conservative escalation, and abstention. CDS-ASR is not proposed for automatic punitive action, account restriction, guilt determination, or direct law-enforcement reporting without human review.

A Appendix: Aggregate Artifacts And Submission Notes

The appendix records the aggregate artifacts used by the manuscript. Long operation records, validation gate commands, candidate-lane details, and full claim registries are moved out of the main text to keep Paper 4-a speech-centered.

- ASR split comparison: `70_experiments/runs/janus_258_test_split_asr_cds_proxy/asr_cds_proxy_comparison.tsv`
- Predictor comparison: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_comparison.tsv`
- Predictor clustered CI: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_clustered_ci.tsv`
- Recovery replay: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/policy_comparison.tsv`
- Fixed-budget frontier: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/fixed_budget_recovery_frontier.tsv`
- Risk atom review: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_risk_atom_review.tsv`

- Completed dual-reviewer audit record: `70_experiments/runs/janus_300_high_stakes_human_audit_completed_300_dual_reviewer_2026_06_02/`
- CEIS ablation summary: `70_experiments/runs/janus_300_high_stakes_ceis_ablation_dual_reviewer_2026_06_02/`
- CEIS method summary: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/ceis_method_summary.tsv`
- Artifact manifest: `80_semantic_risk_asr/paper/artifact_manifest.tsv`

B Ablation Status

Aggregate-safe CEIS ablation is complete on the selected-300 dual-reviewer surface. The tracked output folder contains `ceis_ablation_summary.json`, `ceis_ablation_predictor_summary.tsv`, `ceis_ablation_policy_replay.tsv`, `ceis_ablation_model_summary.tsv`, `ceis_ablation_delta_summary.tsv`, and an interpretation note. These outputs are aggregate-only and exclude transcript-bearing reviewer sheets, merged sheets, raw audio, raw transcripts, row identifiers, reviewer notes, and completed review packages.

References

- [1] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970.
- [2] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.